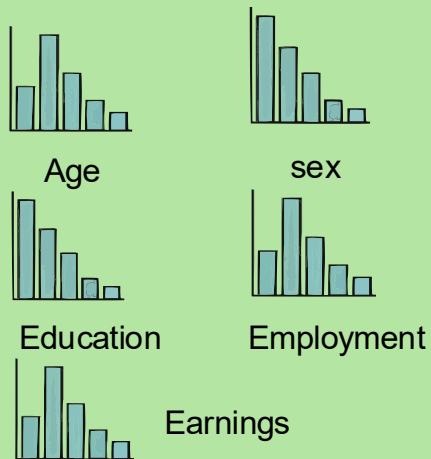
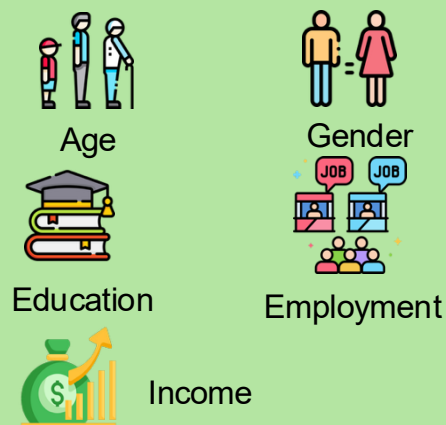


Step1: Regional Targets and External Conditions

External Census Summaries



PUMS Microdata



Encode

Condition
Encoder

Step2: Regional-Context Diffusion Training

Predict Noise

Joint Diffusion
Model

4-attribute Joint Target

Regional
Context

Person-level
earnings model

Conditional Earnings Target

Step3: Synthetic Population Generation

4-attribute Regional Joint Distribution

Generate

Sample

Assign Earnings

5-attribute Synthetic Population

Age
Sex
Education
Employment
Earnings